

University of Macau
CISC7021 – **Applied Natural Language Processing**
Assignment 2, 2024/2025
(Due date: **30 September 2024**)

Introduction

In this assignment, we will utilize a toy Llama-2 (Touvron et al., 2023) language model (the same model used in Assignment 1) to conduct a sentiment analysis task using **zero-shot prompting** and **fine-tuning**, and subsequently evaluate its performance. You will learn how to prompt the language model for sentiment classification, evaluate its task-specific performance, and perform fine-tuning using the PyTorch¹ and HuggingFace libraries.

Skeleton Code and Environment Setup

Most of the code is provided to help you get started quickly. You can reuse or adjust it according to your needs. Necessary tutorials and explanations of function parameters are included in the notebook file.

Free GPU resources, such as Colab and Kaggle, have quotas. To maximize these resources, you can begin by debugging your code on the CPU, and then run it on the GPU to take full advantage of the available free GPU quotas. You are also welcome to use your personal GPU cards for this assignment, if available.

***Note:** In this assignment, the prompting and evaluation tasks (Tasks 1 and 2) can be performed quickly on the CPU!*

Base Language Model

The base language model is a toy model pre-trained on the TinyStories (Eldan et al., 2023) dataset using the Llama-2 architecture. Specifically, this base model was pre-trained on a synthetic dataset of short stories that contain words typically understood by 3 to 4-year-olds. Although this model is much smaller than Llama-2² and has only 42 million parameters, it can still produce a diverse set of fluent and consistent stories. We use this model to help you understand the prompting and fine-tuning process of pre-trained language model.

Data

We used Rotten Tomatoes movie review (Pang et al., 2005) and SST-5 (Socher et al., 2013) datasets. Rotten Tomatoes dataset is split into training (8,530), dev (1,066) and test sets (1,066). SST-5 refers to the Stanford Sentiment Treebank dataset, which

¹ <https://pytorch.org/>

² Even the smallest Llama-2 model has 7 billion parameters.

consists of five labels at the sentence level. The dataset is divided into training (8,544), development (1,101), and test sets (2,210).

Tasks

Detailed instructions of each task can be found in the attached notebook file.

1. Use `prompting_classification()` function to perform zero-shot prompting for sentiment analysis using the Rotten Tomatoes and SST-5 datasets on both the development (validation) and test sets.
 - (i) You are encouraged to experiment with different prompts and discuss your findings.
2. Use functions to evaluate the accuracy and F1 score. The evaluation function for accuracy has already been implemented.
 - (i) You are required to complete the remaining code in the `calculate_f1_score()` function.
 - (ii) Following that, you need to report the classification performance for the experimental results on Task 1.
3. Use `trainer.train()` function to perform fine-tuning on the training data split for both the Rotten Tomatoes and SST-5 datasets.
 - (i) You need to evaluate the classification performance of the fine-tuned model on development (validation) and test splits for both datasets.
 - (ii) You are encouraged to add additional dataset, implement data cleaning techniques, and experiment with different fine-tuning hyperparameters. Be sure to discuss any interesting findings in detail.
4. (Optional) Design different architectures for the classifier and tune the hyperparameters to achieve better performance.
5. (Optional) Consider exploring other parameter-efficient fine-tuning methods that may speedup fine-tuning, such as tuning only the layers that comprise the classifier.

Submission

Notebook and Result Files (20%) Please submit a Jupyter Notebook containing your code and experiments. Ensure that you record the results within your environment. Additionally, make sure to include the result files produced by the provided functions:

- `rto-test-finetuned.txt`
- `rto-test-prompting.txt`
- `rto-val-finetuned.txt`
- `rto-val-prompting.txt`

- sst5-test-finetuned.txt
- sst5-test-prompting.txt
- sst5-val-finetuned.txt
- sst5-val-prompting.txt
- task1-scores.txt
- task3-scores.txt

Report (80%) Submit a 2-4 pages report of your work. (1) Clearly present your experiments, achievements, and solutions to any problems you encountered during the process. (2) Report the classification performance of different methods/models in terms of accuracy, F1 score, and other relevant metrics. (3) Discuss your findings regarding your exploration of possible factors that may affect prompting and fine-tuning performance. (4) Provide an analysis of classification errors. This can include case studies or any bottlenecks for improvement that you identified. The report should follow the ACL proceeding format.³

Use of Generative AI

Please attach an additional section named “Use of Generative AI” at the end of your report. Whether it’s for research, editing or just getting those creative juices flowing, let us know how you’ve used it in your assignments. At University of Macau, all submissions, written or otherwise, should be accompanied with any or all of the following statements, as appropriate.⁴

Declaration Example:

I acknowledge the use of [name of AI tool(s) and hyperlink] to help me [...].

I declare that [content...] generated by AI has been presented and submitted as my own work.

Acknowledgements

The base model checkpoint was converted from llama2.c⁵ project.

References

- [1] Eldan, R., & Li, Y. (2023). Tinstories: How small can language models be and still speak coherent english? *arXiv preprint arXiv:2305.07759*.
- [2] Pang, B., & Lee, L. (2005). Seeing Stars: Exploiting Class Relationships for Sentiment Categorization with Respect to Rating Scales. In *Proceedings of the 43rd Annual Meeting of the Association for Computational Linguistics (ACL’05)*.

³ <https://github.com/acl-org/acl-style-files>

⁴ <https://ctle.um.edu.mo/resource/learning-with-generative-ai-at-um-module/>

⁵ <https://github.com/karpathy/llama2.c>

pp.115-124.

- [3] Socher, R., Perelygin, A., Wu, J., Chuang, J., Manning, C. D., Ng, A. Y., & Potts, C. (2013). Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 conference on empirical methods in natural language processing*. pp.1631-1642.
- [4] Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., . . . Bhosale, S. (2023). Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*.